GeoBEACON: Joint Semantic, Streaming, and Lighting Budget Control for a Vulkan Urban Digital Twin

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Abstract

Urban digital twins must distribute finite frame time, graphics memory, and upload bandwidth across geometry detail and lighting accuracy. Fixed level of detail (LOD) and distance-only selection ignore the task relevance of landmarks, intersections, and route corridors; lighting-only adaptation ignores the cost of making city geometry resident. This paper presents GeoBEACON, a reproducible Vulkan research renderer over a checked OpenStreetMap (OSM) extract of Connaught Place, New Delhi. GeoBEACON combines semantic tile utility, asynchronous three-level geometry streaming, camera-space clustered lighting, aggregate-bounded diffuse-light pruning, temporal hysteresis, and a 10 Hz budget controller. The artifact includes deterministic OSM preprocessing to glTF 2.0 and 3D Tiles 1.1, five recorded camera paths, dual-driver measurement classes, exact diffuse captures, raw frame CSVs, and figure regeneration. A 20-case dual-driver pilot completed without failures. On MoltenVK, bounded GeoBEACON reduced median measured cluster-plus-lighting GPU time from 0.3086 ms for fixed LOD1/fixed clusters to 0.2621 ms in this 100-light validation workload; the result is reported as artifact evidence rather than a general performance claim.

1 Introduction

An urban renderer has coupled resource decisions. Selecting a detailed facade increases vertex and index traffic, resident memory, upload work, visible primitives, and potentially the number of fragments that traverse local light lists. Conversely, aggressive geometry simplification can erase the landmarks and intersections that make a digital twin useful even when conventional image error is small. A controller that treats geometry, streaming, and lighting as independent subsystems can therefore satisfy each local budget while making a poor global allocation.

GeoBEACON tests the following hypothesis:

> *Joint semantic and view-aware allocation of geometry detail, streaming bandwidth, memory, and lighting accuracy provides greater perceptual and task-relevant utility per millisecond than fixed LOD, distance-only LOD, or lighting-only adaptation.*

The contribution is the unified and ablatable allocation system, not SSBOs, clustered shading, instancing, glTF, 3D Tiles, or bounded light selection individually. The implementation retains the original BEACON techniques as regression baselines and adds five city policies: fixed LOD1, distance LOD, semantic LOD, GeoBEACON exact, and GeoBEACON bounded.

The artifact makes three distinctions that are essential for credible systems research. First, conservative omitted-light bounds and measured image error are separate quantities. Second, Vulkan GPU timestamps, Vulkan CPU fallback timings, and analytical predictions occupy separate result classes. Third, adaptation mode includes controller settling, while steady-state experiments can freeze parameters after warm-up.

2 Prior Work and Positioning

Tiled and clustered shading established spatially restricted light lists as a practical forward many-light technique. Lightcuts and later virtual-point-light methods established bounded approximation, while grid-based reservoirs such as ReGIR combine spatial grids with stochastic reuse for much larger light populations [1, 2]. GeoBEACON remains deterministic and targets finite-radius direct point lights. Neural visibility caches introduce online learning and inference behavior outside this study's reproducibility constraints [3].

Geospatial runtimes commonly use hierarchical tiles, screen-space error, and progressive content. OGC 3D Tiles 1.1 standardizes a spatial hierarchy over renderable tile payloads [4]. OSM provides open vector geometry and semantic tags under the ODbL [5]. GeoBEACON connects these standards-oriented data structures to a renderer-level joint controller and exposes semantic weighting as a dedicated ablation.

3 Canonical Dataset

The study area is Connaught Place, New Delhi, bounded by latitude 28.6270–28.6365 and longitude 77.2070–77.2245. The repository commits the canonical OSM 0.6 XML extract so building and benchmark execution do not depend on network availability. The source SHA-256 is recorded in every derived manifest. Network acquisition is an explicit offline action with one-request concurrency, identifying user agent, cache, size limit, timeout, and exponential backoff.

The deterministic preprocessor parses nodes, ways, and multipolygon relations; rejects malformed features; converts WGS84 coordinates to a double-precision local east-north frame; and partitions features into 128 m cells. Height precedence is explicit:

$$h = \begin{cases} \texttt{height}, & \text{when valid,} \\ 3.2\,\texttt{building:levels}, & \text{otherwise,} \\ h_{\text{class}}, & \text{otherwise.} \end{cases}$$

Minimum levels, bridges, tunnels, and layer tags influence vertical placement. Roads become class- and lane-aware ribbons. Materials use deterministic colors rather than downloaded imagery. The checked output contains 151 spatial records and 453 GLBs across LOD0, LOD1, and LOD2. Every GLB is reloaded before acceptance, and the report records vertex/index counts, invalid inputs, semantic counts, bounds, checksums, attribution, and license.

LOD0 contains tile ground, urban proxies, and major roads. LOD1 contains extruded footprints, roads, water, vegetation, and semantic colors. LOD2 retains individual features and minor roads. The runtime manifest accompanies an OGC-style 3D Tiles 1.1 `tileset.json`. Engine code remains MIT licensed; source and derived OSM data carry separate ODbL notices. The rendered window includes "© OpenStreetMap contributors".

4 Streaming Runtime

The geospatial runtime is independent of the tutorial's object map. Each tile progresses through `Unloaded`, `Queued`, CPU loading, upload waiting, `Resident`, eviction, or `Failed`. Worker threads perform file I/O and GLB decoding only. Vulkan model creation occurs on the render thread, preserving device ownership. A generation counter cancels stale requests when the desired LOD changes before upload.

At 10 Hz, the selector performs double-precision bounds tests and ranks visible candidates. For a tile t and representation r, utility is

$$U(t, r) = w_s(t) A_{\text{screen}}(t) V(t) Q(r) S_{\text{temporal}}(t), \quad (1)$$

and cost is

$$C(t, r) = C_{\text{gpu}}(t, r) + \lambda_m M(t, r) + \lambda_b B(t, r) + \lambda_l L(t, r). \quad (2)$$

Here w_s is semantic importance, A_{screen} is projected importance, V is visibility confidence, Q is representation quality, S_{temporal} rewards stability, M is memory, B is upload demand, and L estimates lighting work. Candidates are ordered by marginal utility per byte for deterministic selection.

Visible landmarks, primary roads, intersections, and route-adjacent content receive larger importance. Fixed LOD1 disables semantic differentiation; distance LOD removes w_s; semantic LOD uses w_s without lighting control. GeoBEACON exact and bounded use the joint policy. Representation downgrades require 30 controller ticks of lifetime, and at most eight tile changes enter a frame. Separate upgrade/downgrade pressure prevents oscillation.

Upload bandwidth is enforced by a token bucket. Cold-cache runs begin with no resident models; warm-cache runs bypass bandwidth throttling during the warm-up residency phase. Resident bytes are recomputed from actual representation files. Budget-aware LRU eviction excludes visible tiles and moves resources into a deferred retirement queue. Retirement occurs only after more frames than the maximum frames in flight and a device-idle synchronization before destruction, preventing in-flight resource release. Failed content remains a recorded diagnostic state rather than silently disappearing.

The runtime reports visible, requested, resident, upgraded, downgraded, and failed tiles; bytes resident and uploaded; p95 request latency; representation churn; semantic utility; and budget violations. These are frame-level values, not only end-of-run aggregates.

5 Camera-Space Lighting

The exact diffuse reference stores an unrestricted light array in an SSBO and evaluates all active finite-radius point lights. The visual forward baseline retains specular shading but is excluded from formal error comparisons. A light's diffuse contribution uses finite-radius attenuation and inverse squared distance.

The fixed GPU builder uses a $16 \times 9 \times 24$ camera-space grid. XY membership comes from the projected light sphere, while logarithmic depth is

$$z_{\text{slice}} = \left\lfloor N_z \frac{\log(z/z_n)}{\log(z_f/z_n)} \right\rfloor . \tag{3}$$

Assignment consists of five compute stages:

1. clear headers, cursors, and block sums;
2. dispatch one invocation per light and atomically count its cluster range;
3. perform 256-element block exclusive scans and scan block sums;
4. add block offsets and record overflow;
5. scatter light IDs into compact contiguous storage.

Compute barriers establish count-to-scan, scan-to-offset, offset-to-scatter, and scatter-to-fragment visibility. Timestamp labels isolate count, local scan, block scan, offset, and scatter intervals.

No overflow is silently truncated. A header whose candidate count exceeds its list cap, or whose offset would exceed storage, sets an overflow flag. The fragment shader then evaluates all global lights for that cluster, which is slower but exact because finite-radius attenuation rejects non-influential lights.

5.1 Adaptive bounded evaluation

Adaptive BEACON starts with coarse XY tiles and chooses 1, 2, 4, or 8 depth leaves. Exact mode keeps all intersecting lights. For bounded mode, light l and cluster c receive the diffuse bound

$$E_{\max}(l, c) \leq \frac{I_l \rho_{\max}}{\max(d_{\min}(l, c)^2, \epsilon_d)} . \tag{4}$$

Candidates are sorted or bucketed from weakest to strongest. Removal stops before the aggregate omitted bound exceeds ε_c:

$$\sum_{l \in R(c)} E_{\max}(l, c) \leq \varepsilon_c . \tag{5}$$

This prevents many individually weak lights from collectively violating quality. Sparse clusters store explicit 32-bit indices. Dense clusters store bitsets. Bitset traversal is word-wise: `findLSB` selects a set bit and `word &= word-1` clears it, so shader work scales with retained lights rather than the global light population.

6 Joint Budget Controller

The controller receives target frame milliseconds, memory bytes, upload MiB/s, global diffuse error, and a maximum tile-change rate. Measured cluster-build, lighting, upload, and total-frame values enter exponentially weighted moving averages. Scheduling occurs at 10 Hz rather than every frame. This separates resource decisions from frame noise and reduces list and residency churn.

Geometry and lighting share priorities. Landmark, intersection, primary-road, and near-facade tiles receive tighter local light-error allowances. Distant and low-importance tiles receive looser allowances, but the controller never exceeds the global sum of omitted bounds. When total frame time exceeds the high threshold, the controller reduces changes and subdivision pressure. When time remains below the low threshold, it cautiously improves quality. The split and merge thresholds are different, and minimum lifetimes prevent threshold chatter.

The controller exposes its decisions through counters rather than hiding them in a black box: subdivision depth, active clusters, explicit/bitset distribution, retained and pruned samples, predicted bound, split/merge count, cluster churn, visible/resident tiles, LOD churn, uploads, memory, and budget violations. This transparency supports ablation and replay.

7 Experimental Protocol

Five deterministic camera paths target different failure modes: outer orbit tests broad visibility; street drive tests continuity; intersection dwell emphasizes semantic hotspots; landmark approach tests progressive quality; and rapid teleport stresses cancellation, upload, and eviction. Benchmark frames use a fixed 1/60 s timestep and recorded transforms.

The full matrix spans 100, 500, and 2,000 lights; 16.67 and 33.33 ms frame budgets; 256, 512, and 1,024 MiB memory; 25, 100, and effectively unlimited MiB/s upload; 720p, 1080p, and 1440p; cold and warm caches; MoltenVK and KosmicKrisp; and five repeated trials. Each full case warms for 600 frames and measures 1,800 frames.

Every run records the git commit, shader hashes, device identity and UUID, Vulkan API and driver, capabilities, policy, path, random seed, resolution, cache mode, source checksum, budgets, and measurement class. GPU-query rows are never merged with CPU fallback or analytical predictions. The exact reference uses maximum resident geometry for its selected frame, exact diffuse SSBO lighting, identical camera/light state and exposure, and no light pruning.

8 Pilot Results

The checked pilot is an instrumentation validation: 20 cases, all five policies, two routes, both local ICDs, 100 lights, 640×360, 256 MiB, 25 MiB/s, cold cache, 30 warm-up frames, and 60 measured frames. All 20 completed. It is intentionally reported separately from the full five-repeat matrix.

Table 1: Pilot medians and p95 CPU frame time. GPU is cluster build plus lighting.

Driver	Policy	CPU50 ms	CPU95 ms	GPU50 ms
MoltenVK	Fixed LOD1	8.232	11.920	0.3086
MoltenVK	Geo exact	8.108	12.651	0.2866
MoltenVK	Geo bounded	7.915	12.361	0.2621
KosmicKrisp	Fixed LOD1	16.562	25.909	–
KosmicKrisp	Geo exact	16.046	21.421	–
KosmicKrisp	Geo bounded	16.105	21.838	–

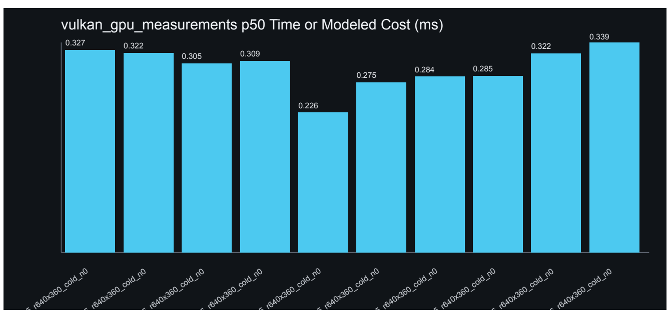

Figure 1: MoltenVK timestamp-query pilot values generated from raw frame CSVs.

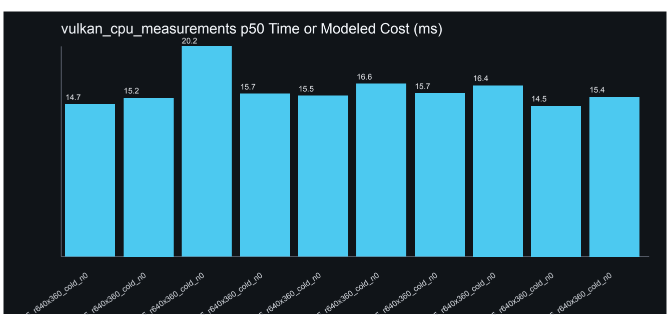

Figure 2: KosmicKrisp CPU-fallback pilot values; no GPU time is inferred.

MoltenVK reported timestamp queries; KosmicKrisp did not. The bounded policy's lower GPU median in this workload is consistent with fewer evaluated light samples, but the pilot is too small for a cross-scene causal claim. CPU p95 increased for adaptive policies because CPU hierarchy construction and exact capture synchronization remain visible in the validation configuration. This negative result is retained.

Every policy uses a separately streamed maximum-LOD diffuse reference. The saved PPM captures verify that the compared buffers contain rendered city geometry rather than only a clear color. Mean MSE was 3.09×10^{-4} for fixed LOD1 and approximately 5.11×10^{-4} for semantic adaptive policies on MoltenVK. Geometry allocation dominates these values in the pilot, while the conservative omitted-light bound remains a separate reported quantity.

9 Verification and Reproduction

CTest covers contribution bounds, encoding selection, controller response, exclusive scan offsets, capacity overflow, coordinate conversion, height precedence, polygon orientation, source checksum, and reload of all 453 committed GLBs. Release builds compile vertex, fragment, and compute shaders as executable dependencies. Physical-device selection accepts index, UUID, or name substring. Instance creation negotiates Vulkan 1.2 against the loader and enables portability enumeration when available.

Both local ICD manifests execute:

- `/usr/local/share/vulkan/icd.d/MoltenVK_icd.json`;
- `/usr/local/share/vulkan/icd.d/libkosmickrisp_icd.json`.

The matrix runner is resumable: a case with an existing frame CSV is treated as cached unless `--force` is supplied. It writes a matrix manifest including failures and elapsed time. The figure script uses only the Python standard library and regenerates SVG plots and a Markdown summary from raw rows.

Correctness invariants are explicit. Exact modes retain every intersecting light. A list overflow selects the exact fallback. Worker threads never call Vulkan. Stale loads are discarded by generation. Resident resources enter deferred retirement and are synchronized before destruction. OSM source acquisition never occurs at startup. Every render window displays source attribution.

10 Research Scope

The formal light bound applies to opaque, unshadowed diffuse point lighting with finite radius. Specular shading remains a visual baseline and is not claimed to be strictly bounded. Shadows, transparency, neural rendering, street imagery, Gaussian splats, sensor simulation, ROS, CARLA, and live network streaming are outside the evaluated claim. These are scope boundaries, not hidden missing result classes.

The canonical city remains within a compact local coordinate domain, while selection and manifest bounds use double precision. The renderer preserves a floating-origin-compatible data model; the checked paths do not exceed the domain where rebasing affects float precision. Performance values are hardware-local, while build, correctness, preprocessing, and analytical tests are portable.

11 Conclusion

GeoBEACON turns a tutorial renderer into a reproducible urban rendering research artifact. It joins real OSM ingestion, deterministic tiled geometry, semantic utility, asynchronous residency, camera-space count/scan/scatter clustering, error-bounded diffuse pruning, temporal stability, budget feedback, exact image capture, raw measurements, and generated figures under one ablatable interface.

The pilot establishes that all five policies and both local Vulkan drivers execute over real Connaught Place geometry, that timing classes remain separated, and that exact/bounded image comparisons operate on saved rendered frames. It also exposes a meaningful trade-off: bounded lighting reduced measured MoltenVK cluster-plus-lighting time in the pilot, while adaptive CPU work raised frame-time tails. The full matrix is encoded as a deterministic runner so larger empirical claims can be regenerated without changing the renderer or measurement definitions.

References

[1] M. MacDonald et al., “Sparse Sampling and Completion for Light Transport in VPL-based Rendering,” arXiv:2202.12567, 2022.

[2] J. Boksansky et al., “Rendering Many Lights with Grid-Based Reservoirs,” NVIDIA Real-Time Graphics Research, 2021.

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[4] Open Geospatial Consortium, “3D Tiles Specification 1.1,” OGC 22-025r4, 2023.

[5] OpenStreetMap Foundation, “Copyright and License,” Open Database License 1.0.